
STA5076Z – Supervised Learning 2026

Assignment 3

Due Date: Wednesday 27 May 2026, 23:59

Please read the following instructions carefully:

- Answer all questions using R; you may use any packages and functions you like.
 - Present your work as a **video presentation**, which is to be submitted to **Assignment 3 – Presentation** on Amathuba. Any format is acceptable. Submit any visual aid used in the video in the same location.
 - The assignment provides you the opportunity to present your statistical work in another format, one that is often relevant in the workplace. As such, you will be marked on your ability to convey the task in a concise and professional manner.
 - Upload all relevant R files (.R, .Rmd, .qmd, .Rdata, etc.) as well as your test set predictions to **Assignment 3 – Code & Predictions**. **Please do not zip these files**; upload them separately.
 - Set seeds appropriately to ensure that all results are exactly reproducible.
 - You are encouraged to include figures and tables, but only if they are relevant to your discussion. Use descriptive captions! One should be able to understand what a figure represents or a table contains based on the caption.
 - Also be sure to interpret figures sufficiently; marks will be deducted for any output placed in the presentation without discussion.
 - Structure your presentation according to the description given below.
 - Reference your sources appropriately.
 - **The video is limited to 8 minutes**. This is the limit, not the target. If you can comprehensively tell your story in less time, then well done!
 - You are expected to work on this **on your own**.
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Cape Town Airbnb listing analysis

Extract from Assignment 2:

You are provided with `listings.csv`, containing (curated and filtered) Airbnb listing data for Cape Town from 2025. This file must be read into R as is, i.e. the spreadsheet may not be edited at all. The feature descriptions are given in `variable_descriptions.csv`. The target variable is `fully_booked_30`, which indicates whether or not a listing had zero availability over the next 30 days at time of data collection. **The positive/target class to use throughout this exercise is ‘no’**, i.e. we are interested in identifying properties that are currently expected to be vacant for at least one night over the next month.

The data have already been cleaned and are ready for modelling. Although an extensive exploratory data analysis is not required, a brief section on this may be included. However, if you do so, your discussion should focus only on aspects that pertain to this exercise. You do not *have* to use all the features for the modelling, but if you omit any, this must be clearly communicated. If you want to (not necessary), you may perform additional preprocessing and/or feature engineering, as long as you **clearly document and report** what you did.

You are also provided with an unlabeled test set (`testing.csv`) for which predictions are to be uploaded.

Presentation Description

Create a scenario which motivates the study of the Cape Town Airbnb listing data. This scenario will provide discussion guidelines throughout the presentation. For example, you could be a consultant for the municipality trying to understand the depth of the Airbnb issue.

Under your scenario, provide an introduction and data exploration. This will likely be very similar to assignment two; however, your illustrations may differ given the new format.

Using the full training set, fit the following models.

1. An SVM with a *linear kernel*.
2. A nonlinear kernel SVM.
3. A feed-forward neural network.

You should present a high-level recap of the models and their mechanics. These should be integrated into a discussion on your modelling choices.

For example, when you tune the SVM models, you can intuitively explain what the *cost* parameter does and how you have chosen it. Clearly, you need to be wary of time. No one concept is absolutely required for discussion, the flow of your discussion is the key.

Present a model evaluation and comparison. You have some freedom in how you illustrate your results, but the metrics and figures you choose should support your final model selection. You should motivate your final model in relation to the scenario introduced in Section 1.

Potential evaluation metrics to consider include Accuracy, F1 Score, Recall, Precision, AU-ROC, AUPRC. Once again, the metrics you present should be accompanied by discussion.

If you use any metrics that require a predicted label, motivate an appropriate decision rule.

Prediction Submission

This section does not form part of the video presentation.

Use your selected model to predict the `fully_booked_30` probabilities for the observations in `testing.csv`. Write your predictions to a .csv file as follows:

- The file must only contain ONE column, namely the predicted probabilities, i.e. just a column of numbers between 0 and 1.
- Remember, these must represent $\hat{\mathbb{P}}(\text{fully_booked_30} = \text{'no'})$.
- Do NOT include a column header.
- There should be NO blank cells to the left or above this column.
- The order of the predictions must correspond to the order of the observations in `testing.csv`.
- The file name must be your student ID, all uppercase. E.g. `STDNUM001.csv`.

$\mathcal{END}$

TOTAL MARKS = 100
